Self-Check

Lecture 0

Prerequisites: A Refresher

This diagnostic checks the four computational skills the early lectures rely on. Take it *cold*, without notes, before Lecture 1; then check your work and read the routing guide at the end to see what (if anything) to revisit in `lecture-0.pdf`. Throughout, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$. Tags in parentheses indicate the flavour of a problem.

Complex arithmetic

Exercise 0.1 (computational) Let $z = 3 + 2i$ and $w = 1 + i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Exercise 0.2 (computational) Write $1 - i$ in polar form and use it to compute $(1 - i)^8$.

Exercise 0.3 (computational) Find all complex numbers z with $z^2 = i$.

Exercise 0.4 (computational) Let $z = 3 + i$ and $w = 1 + 2i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Exercise 0.5 (computational) Write $1 + i$ in polar form and use it to compute $(1 + i)^6$.

Exercise 0.6 (computational) Write $\sqrt{3} + i$ in polar form and use it to compute $(\sqrt{3} + i)^4$, giving the answer in Cartesian form.

Exercise 0.7 (proof) For complex numbers $z = a + bi$ and $w = c + di$, prove that $\overline{zw} = \overline{z}\,\overline{w}$.

Exercise 0.8 ($\star$) Find all four complex numbers z with $z^4 = -4$.

Matrix algebra

Exercise 0.9 (computational) Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. Compute AB, BA, and A^T, and verify $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ even though $AB \neq BA$.

Exercise 0.10 (computational) Let $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$. Compute AB and BA, and verify $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ although $AB \neq BA$.

Exercise 0.11 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, compute A^2 and A^3.

Exercise 0.12 (computational) Let $A = \begin{pmatrix} 1+i & 2-i \\ 0 & 3i \end{pmatrix}$ over $\mathbb{C}$. Write down the transpose A^T and the conjugate transpose $A^\dagger = \overline{A}^\mathsf{T}$.

Exercise 0.13 (proof) For square matrices A, B of the same size, prove the cyclic identity $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.

Exercise 0.14 (proof) For matrices $A \in \mathcal{M}_{m\times n}(\mathbb{F})$ and $B \in \mathcal{M}_{n\times p}(\mathbb{F})$, prove that $(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}$.

Determinants

Exercise 0.15 (computational) Evaluate $\det\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 5 & 6 & 0\end{pmatrix}$ and state whether the matrix is invertible.

Exercise 0.16 (computational) Use the 2×2 formula to invert $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$.

Exercise 0.17 (computational) For $A = \begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$, find all λ for which $\det(\lambda I - A) = 0$. (These values reappear in Week 5 as the eigenvalues of A.)

Exercise 0.18 (computational) Evaluate $\det\begin{pmatrix}1 & 2 & 1\\ 0 & 3 & 2\\ 1 & 0 & 1\end{pmatrix}$ and state whether the matrix is invertible.

Exercise 0.19 (computational) Let $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B = \begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}$. Compute $\det A$, $\det B$, and $\det(AB)$, and confirm $\det(AB) = \det A\ \det B$.

Exercise 0.20 (computational) Evaluate the block-diagonal determinant $\det\begin{pmatrix}1 & 1 & 0 & 0\\ 1 & 2 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 1 & 3\end{pmatrix}$.

Exercise 0.21 (proof) For 2×2 matrices A, B, prove that $\det(AB) = \det A\ \det B$ directly from the entries.

Gaussian elimination

Exercise 0.22 (computational) Solve, by row reduction,

$$\begin{aligned} x + y + z &= 2,\\ x - y + 2z &= 5,\\ 2x + y - z &= 2. \end{aligned}$$

Exercise 0.23 (computational) Find a basis for the solution space (null space) of $Ax = 0$, where $A = \begin{pmatrix}1 & -1 & 2 & 0\\ 2 & -2 & 3 & 1\end{pmatrix}$, and state its dimension.

Exercise 0.24 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix}1 & 1\\ 1 & 2\end{pmatrix}$.

Exercise 0.25 (computational) Solve, by row reduction,

$$\begin{aligned} x + y + z &= 6,\\ x - y + z &= 2,\\ 2x + y - z &= 1. \end{aligned}$$

Exercise 0.26 (computational) Describe the full solution set of

$$\begin{aligned} x + 2y - \phantom{y+{}} z &= 3, \\ 2x + y + \phantom{y+{}} z &= 3, \end{aligned}$$

in the form "particular solution + span of the null space".

Exercise 0.27 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Exercise 0.28 (proof) Let $A \in \mathcal{M}_{m\times n}(\mathbb{F})$. Prove that the null space $\operatorname{null} A = \{x \in \mathbb{F}^n : Ax = 0\}$ is closed under addition and scalar multiplication (so it is a subspace of $\mathbb{F}^n$).

Exercise 0.29 ($\star$) For which real values of a does the system

$$\begin{aligned} ax + y + z &= 1, \\ x + ay + z &= 1, \\ x + y + az &= 1 \end{aligned}$$

have a unique solution, infinitely many solutions, or no solution? Give the solution in the unique case.

Reading your results

Use the groups above to locate any rust, and the matching section of `lecture-0.pdf` to clear it before Lecture 1.

- **Complex arithmetic** (problems 0.1–0.7). Used on the first pages of Lecture 1 and heavily from Week 7 onward (inner products, Pauli matrices, quantum phases). Any slip here matters most—revisit *Refresher* §1.
- **Matrix algebra** (0.8–0.13). Multiplication, transpose, and trace run through the whole course; the conjugate transpose becomes the adjoint in Week 9. Revisit *Refresher* §2.
- **Determinants** (0.14–0.20). Behind the invertibility test from Lecture 1 and the characteristic polynomial of Week 5—problem 0.16 is a direct preview. Revisit *Refresher* §3.
- **Gaussian elimination** (0.21–0.28). The engine for bases, rank, and dimension in Weeks 2–4, and for the rank–nullity theorem of Week 3. Revisit *Refresher* §4.